Beyond Action-Blind Prediction: Action-Sensitive Driving Video Generation with Diffusion Transformers

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Abstract

Pretrained video diffusion transformers can produce plausible driving videos from visual context alone, but this does not make them controllable world models. We study front-camera future generation: given 49 observed frames at 24 FPS, synthesize the next 72 frames while conditioning on future ego-action signals. The central failure mode is that naive action conditioning, including global action tokens, temporal action tokens, and AdaLN-style action modulation, either gets ignored or damages the visual prior, producing blur, or motion collapse. We present a low-frequency action-conditioning objective for a frozen pretrained diffusion transformer with trainable LoRA and action modules. The final model uses per-frame ego-action vectors containing 112 normalized driving-control features, including motion, velocity, acceleration, and trajectory-derived quantities, together with a temporal action bottleneck, low-frequency future-token target and delta losses, and a high-frequency teacher preservation term. On 1,904 validation windows, the no-action visual baseline has exactly zero action sensitivity by construction, while our selected action model reaches $S_{\text{rgb}} = 2.262$*, meaning its generated future changes measurably when the same visual context is paired with zero, shuffled, or reversed future actions. It preserves near-baseline visual quality: PSNR, a pixel-fidelity metric, changes from 17.153 to 17.180; SSIM, a structural-similarity metric, changes from 0.70789 to 0.70816; and FVD-style distance, a distributional video-quality metric where lower is better, changes only from 11.702 to 11.747. On high-action clips, where braking, acceleration, or turning makes the future less determined by visual context alone, correct actions are consistently favored over wrong action sequences under both pixel and structural metrics, providing evidence of correct-action alignment where future actions matter.*

1. Introduction

Autonomous-driving video prediction is often evaluated as visual future continuation: given past camera frames, predict plausible future frames. This is useful, but it is not sufficient for a driving world model. A world model should answer counterfactual questions: if the same observed scene is followed by a different ego-action sequence, should the generated future change accordingly?

This distinction matters because front-camera driving scenes are strongly constrained by visual context. A model with no access to actions can still produce plausible futures and obtain competitive PSNR, SSIM, or FVD-style scores. But it is action-blind by construction. Such a model can be a good conditional video prior, but it is not controllable.

We adapt the distilled LTX-Video 2B v0.9.8 model [3] , a pretrained video diffusion transformer, into an action-conditioned front-camera future generator. The input is 49 context frames at 24 FPS; the target is the next 72 frames. We condition on future ego-action sequences aligned to the same 121-frame window. The final action representation is a dense 112D vector per frame.

A key empirical finding is that action conditioning initially fails in two ways: it is either ignored, or it disrupts the pretrained visual pathway. simple ways of injecting actions into a pretrained video diffusion transformer fail. Global action tokens, temporal action tokens, and AdaLN-style modulation can make the model react to actions, but they often corrupt the visual pathway. Some variants even improve PSNR and SSIM by smoothing motion and removing high-frequency detail, which makes pixel metrics look better while the video becomes less useful.

The final method, V4-LF, is built around one hypothesis: ego-actions are primarily low-frequency temporal control signals. They should influence coarse future evolution without directly damaging the visual prior.

We implement this using a temporal action bottleneck, low-frequency future supervision, and teacher preservation to keep the pretrained visual pathway stable.

This work makes three contributions. First, we show that a strong no-action driving video predictor is not a controllable world model, even when it obtains competitive visual metrics. Second, we demonstrate that naive action-conditioning mechanisms can create a metric trap, improving PSNR/SSIM while degrading motion and high-frequency structure. Third and most importantly, we introduce V4-LF, our fourth major action-conditioning design, built around a low-frequency temporal objective that produces measurable counterfactual action sensitivity with near-baseline visual fidelity and a positive correct-action advantage on high-action clips.

2. Related Work

Video diffusion transformers. Diffusion models [5] and latent diffusion [12] provide strong generative priors. Diffusion transformers (DiTs) [11] replace convolutional UNets with scalable token-based architectures, and recent video models extend this design to long spatiotemporal generation [2, 19, 8, 3]. Our work does not train a video generator from scratch. We study whether a pretrained video diffusion transformer can be lightly adapted into an action-conditioned driving predictor using LoRA [7] and external action modules.

Controllable video generation. Prior work injects controls through adapters, ControlNet-like branches [21], dense conditioning streams such as VideoComposer [15], trajectory or camera controls [18], and in-context conditioning. FullDiT2 [4], one of the papers that informed our design, argues that concatenating many condition tokens into a full-attention video DiT is powerful but computationally redundant. This matches our experience: broad action-token exposure can be expressive, but it also risks interfering with visual synthesis. We therefore move away from unrestricted global action tokens toward a bottlenecked latent-time residual.

Auxiliary objectives and preservation losses. Recent video-diffusion training papers show that auxiliary objectives can improve generation when they are aligned with the structure of the problem. Align4Gen [9] aligns intermediate video diffusion features with pretrained vision encoders and emphasizes complementary frequency information. Conditional generative video compression [20] uses multi-granular conditioning, role-aware embeddings, and robustness to missing conditions. Enhance-A-Video [10] studies temporal attention and the tension between temporal consistency and spatial detail. These works motivated our separation of low-frequency control from high-frequency appearance preservation.

Driving world models. Generative driving simulators such as GAIA-1 [6] and DriveDreamer [16] study controllable video generation for autonomous driving. Our scope is narrower: we do not build a complete simulator. We isolate one adaptation question: can a pretrained diffusion transformer become action-sensitive and action-aligned on Waymo-style front-camera continuation with limited trainable parameters?

3. Data

We use Waymo Open Dataset [13] front-camera video windows constructed from 24 FPS interpolated clips. Each example contains 121 frames:

$$x_{0:120} = (x_0, \dots, x_{120}), \qquad a_{0:120} \in \mathbb{R}^{121 \times 112}. \tag{1}$$

The model observes the first 49 frames, $x_{0:48}$, and predicts the next 72 frames, $x_{49:120}$. Training loss is applied only to the future segment. The action tensor is aligned frame-by-frame with the same 121-frame video window, so each video frame x_t has a corresponding ego-action vector a_t. We use 7,992 training windows per epoch and 1,904 validation windows.

The original action metadata is available at 10 FPS, while our video windows are 24 FPS. For each 24 FPS frame, we map it to a fractional 10 FPS source frame and linearly interpolate the action metadata between the two nearest source frames. If one source frame is missing, we fall back to the nearest available metadata frame. This gives a dense per-frame action sequence at the same temporal resolution as the video.

3.1. Ego-Action Representation

The final action-conditioned models use the full continuous ego-action representation. These are not discrete labels such as "brake" or "turn." They are numeric ego-motion signals that describe the vehicle's current state, recent motion history, and short-horizon future trajectory. For each frame t, the 112-dimensional vector is

$$a_t = [p_t^{+x},\, p_t^{+y},\, v_t^{-x},\, v_t^{-y},\, \dot{v}_t^{-x},\, \dot{v}_t^{-y},\, s_t,\, \omega_t,\, u_t^{v},\, u_t^{\omega}], \tag{2}$$

with the following breakdown:

The future position terms describe the ego vehicle's planned relative displacement from the current frame. Increasing future x displacement corresponds to continued forward motion, while smaller increments indicate slowing or braking. Future y displacement captures lateral movement such as turns or lane-change-like motion. The past velocity and acceleration terms provide local motion context before the current frame, while speed, yaw rate, and camera-frame velocity describe the instantaneous ego state.

All action features are normalized using training-set statistics before being passed to the model. The model therefore receives continuous standardized action values, not hand-written semantic commands. Any labels such as "brake," "accelerate," or "turning" are used only for evaluation strata, not for training.

Table 1. Per-frame 112D ego-action vector. The vector contains continuous ego-motion quantities, not discrete labels such as brake or turn.

Component	**Dim.**	**Index range**	**Meaning**
`future_pos_x[20]`	20	0–19	Future ego x displacement samples. Larger increments indicate continued forward motion; smaller increments indicate slowing.
`future_pos_y[20]`	20	20–39	Future ego y displacement samples. Sustained lateral displacement corresponds to turning or lane-change-like motion.
`past_vel_x[16]`	16	40–55	Past longitudinal velocity history.
`past_vel_y[16]`	16	56–71	Past lateral velocity history.
`past_accel_x[16]`	16	72–87	Past longitudinal acceleration history. Negative values are braking/deceleration cues.
`past_accel_y[16]`	16	88–103	Past lateral acceleration history.
`speed_mps`	1	104	Current scalar speed in meters per second.
`approx_yaw_rate_rad_s`	1	105	Current approximate yaw rate.
`camera_velocity_v_x/y/z`	3	106–108	Current camera-frame linear velocity.
`camera_velocity_w_x/y/z`	3	109–111	Current camera-frame angular velocity.
Total	**112**		**Per-frame ego-action vector.**

3.2. Temporal Sampling and Latent Bottleneck

Temporal sampling matters because the video VAE applies aggressive spatiotemporal compression. The latent representation downsamples time by a factor of 8 and space by a factor of 32 in height and width. Thus, a local $8 \times 32 \times 32 \times 3$ video volume is represented by one latent site with 128 channels. If the frame rate is too low, one latent timestep spans too much real time, forcing multiple distinct driving motions into a single latent code.

For this reason, we interpolate Waymo clips to 24 FPS before constructing 121-frame windows. At 24 FPS, one 8-frame latent temporal block corresponds to

$$8/24 \approx 0.33 \text{ seconds}, \tag{3}$$

which preserves enough temporal resolution for driving motion while keeping sequence length and compute manageable. We also tested higher frame-rate settings, but 24 FPS gave the best practical balance between temporal detail, compute cost, and generation quality.

4. Method

4.1. No-action visual baseline

The no-action model adapts the pretrained video diffusion transformer to the Waymo continuation task using visual LoRA only. It is the correct visual baseline because it measures how well the model predicts future video without controllability. It is also the correct action baseline because its counterfactual action sensitivity is exactly zero.

4.2. Final action pathway

The final action model uses full 112D frame actions, but does not concatenate them globally into the text token stream. Instead, an action encoder maps frame actions into

Table 2. Final task setup. All final results use the corrected 24 FPS action-aligned windows and seed 231.

Backbone	Pretrained 2B-parameter video diffusion transformer
Input	49 context frames, front camera, 24 FPS
Target	72 future frames, front camera, 24 FPS
Action	112D per-frame ego-action vector
Train windows	7,992 per epoch
Validation windows	1,904
Frozen modules	VAE, text encoder, base transformer
Trainable modules	LoRA adapters, action encoder, action residuals
Sampling	shifted/log-normal timestep sampling
Inference	image-conditioning noise scale fixed to 0.0

latent-time control vectors:

$$z_{0:120} = E_\theta(a_{0:120}), \qquad z_t \in \mathbb{R}^{d_a}. \tag{4}$$

The frame-level action representation is pooled to the diffusion transformer's latent-time bins using the latent token coordinates. For a video token i with latent time $\tau(i)$, the action residual is broadcast spatially:

$$h_i^{(\ell)} \leftarrow h_i^{(\ell)} + \gamma_\ell P_\ell(z_{\tau(i)}). \tag{5}$$

Here $h_i^{(\ell)}$ is the hidden state at block ℓ, P_ℓ is a learned projection, and γ_ℓ is a learned gate. This design makes actions affect temporal evolution at each latent time while avoiding dense per-spatial-token cross-attention to action tokens.

4.3. Loss Function

The final model is trained with a weighted sum of the standard diffusion objective, low-frequency action-alignment losses, a high-frequency teacher-preservation

Context video $x_{0:48}$ + **future actions** $a_{0:120} \in \mathbb{R}^{121\times 112}$

$\Downarrow$

Frozen pretrained video diffusion transformer with trainable LoRA adapters

Temporal action encoder $\rightarrow$ **latent-time bottleneck** $\rightarrow h_i^{(\ell)} \leftarrow h_i^{(\ell)} + \gamma_\ell P_\ell(z_{\tau(i)})$

$\Downarrow$

Diffusion future prediction with low-frequency action losses and high-frequency teacher preservation

Figure 1. Final action-conditioning design. The action path is constrained to latent-time residuals rather than global condition-token concatenation. This implements the hypothesis that ego-actions should control low-frequency temporal evolution while the pretrained visual pathway preserves appearance.

loss, and small regularizers on the action residual path:

$$\begin{aligned}\mathcal{L} =& 0.25\mathcal{L}_{\text{diff}} + 1.00\mathcal{L}_{\text{lf,target}} + 1.00\mathcal{L}_{\text{lf,delta}} \\ &+ 0.20\mathcal{L}_{\text{hf}} + 0.05\mathcal{L}_{\text{aux}} + 0.002\mathcal{L}_{\text{res}} + 0.002\mathcal{L}_{\text{gate}}.\end{aligned} \tag{6}$$

The loss is designed to separate control from rendering: low-frequency terms supervise action-dependent temporal evolution, while the high-frequency teacher term preserves visual detail from the no-action model.

Let z_0 be the clean latent video, z_t be the noised latent at diffusion timestep t, and let ϵ denote the sampled noise. The denoising network predicts either noise or velocity depending on the pretrained model parameterization; we write the corresponding training target generically as u_t. The student prediction is

$$\hat{u}_\theta = f_\theta(z_t, t, x_{\text{ctx}}, a_{0:120}), \tag{7}$$

where x_{ctx} is the 49-frame visual context and $a_{0:120}$ is the 121-frame action sequence. The diffusion loss is the ordinary denoising loss on future latent tokens:

$$\mathcal{L}_{\text{diff}} = \frac{1}{|\mathcal{F}|}\sum_{i\in\mathcal{F}} \|\hat{u}_{\theta,i} - u_{t,i}\|_2^2, \tag{8}$$

where $\mathcal{F}$ indexes tokens belonging to the 72-frame future segment. This term keeps the action-conditioned student aligned with the pretrained diffusion objective.

The low-frequency target loss operates on spatial averages at each latent time. Let $\tau(i)$ map token i to its latent-time index, and let Ω_τ be the set of spatial tokens at latent time τ. For any token field y_i, define

$$\bar{y}_\tau = \frac{1}{|\Omega_\tau|}\sum_{i\in\Omega_\tau} y_i. \tag{9}$$

We apply this to the student prediction and the diffusion target:

$$\mathcal{L}_{\text{lf,target}} = \frac{1}{|\mathcal{T}_\mathcal{F}|}\sum_{\tau\in\mathcal{T}_\mathcal{F}} \left\|\bar{\hat{u}}_{\theta,\tau} - \bar{u}_{t,\tau}\right\|_2^2, \tag{10}$$

where $\mathcal{T}_\mathcal{F}$ is the set of latent timesteps that overlap the future segment. This loss forces the action branch to match coarse future state rather than only local pixel-level detail.

The low-frequency delta loss matches temporal changes in these spatial averages. Define

$$\Delta\bar{y}_\tau = \bar{y}_{\tau+1} - \bar{y}_\tau. \tag{11}$$

Then

$$\mathcal{L}_{\text{lf,delta}} = \frac{1}{|\mathcal{T}_\mathcal{F}|-1}\sum_{\tau\in\mathcal{T}_\mathcal{F}} \left\|\Delta\bar{\hat{u}}_{\theta,\tau} - \Delta\bar{u}_{t,\tau}\right\|_2^2. \tag{12}$$

This term explicitly supervises coarse temporal evolution. It is the part of the objective most directly tied to action conditioning: acceleration, braking, and turning should primarily affect how the future changes over time.

The high-frequency teacher loss uses the corrected no-action model as a frozen teacher. Let

$$\hat{u}^T = f_T(z_t, t, x_{\text{ctx}}) \tag{13}$$

be the teacher prediction using the same noisy latent and timestep but no action input. For any token field y_i, define its high-frequency residual by subtracting the spatial mean at the same latent time:

$$\text{hf}(y_i) = y_i - \bar{y}_{\tau(i)}. \tag{14}$$

The teacher preservation term is

$$\mathcal{L}_{\text{hf}} = \frac{1}{|\mathcal{F}|}\sum_{i\in\mathcal{F}} \left\|\text{hf}(\hat{u}_{\theta,i}) - \text{hf}(\hat{u}_i^T)\right\|_2^2. \tag{15}$$

This term does not force the student to copy the teacher globally. It only constrains the spatial residual after removing the per-time mean. Therefore the student can still change low-frequency future motion, while being penalized for destroying high-frequency structure such as edges, texture, and local appearance.

The auxiliary action-motion loss, $\mathcal{L}_{\text{aux}}$, is applied to the action branch output before it is injected into the diffusion transformer. Its purpose is to encourage the action encoder to represent temporal motion information rather than collapse to an uninformative residual. In implementation, this term is computed on the low-frequency temporal action representation and is used only as a weak auxiliary signal; its weight is much smaller than the two main low-frequency losses.

The residual regularizer penalizes the magnitude of the action residuals injected into the transformer:

$$\mathcal{L}_{\mathrm{res}} = \frac{1}{|\mathcal{B}|} \sum_{\ell \in \mathcal{B}} \|r_\ell\|_2^2, \tag{16}$$

where r_ℓ is the action residual at transformer block ℓ, and $\mathcal{B}$ is the set of blocks receiving action conditioning. This prevents the action pathway from overpowering the pretrained visual pathway.

Finally, the gate regularizer penalizes the learned scalar action gates:

$$\mathcal{L}_{\mathrm{gate}} = \frac{1}{|\mathcal{B}|} \sum_{\ell \in \mathcal{B}} \gamma_\ell^2. \tag{17}$$

The gates are initialized near zero, so the model begins close to the no-action visual baseline. During training, the gates can grow only when action conditioning improves the objective enough to overcome this penalty.

Overall, the loss separates the two roles of the model. The diffusion and high-frequency teacher terms preserve the pretrained video prior, while the low-frequency target and delta terms make the future trajectory action-dependent. This separation is necessary because direct global action conditioning tended to mix low-frequency control with high-frequency rendering, producing blur or unstable motion.

4.4. How the objective was discovered

The final objective was not chosen a priori. Table 3 summarizes the experimental path. The final loss is only interpretable as a response to earlier failures.

5. Metrics

We report two metric families.

Visual metrics. We compute future-frame PSNR, SSIM [17], FVD-style distance [14], Laplacian sharpness ratio, FFT high-frequency ratio, motion ratio, temporal delta error, boundary error, and copy leakage. PSNR and SSIM are useful but insufficient because smoothed video can score well while losing motion and detail [1].

Action metrics. Counterfactual sensitivity measures whether changing the action sequence changes the generated future:

$$S_{\mathrm{rgb}} = \frac{1}{3} \sum_{a' \in \mathcal{A}_{\mathrm{wrong}}} \mathrm{MAE}\big(G(a_{\mathrm{correct}}), G(a')\big), \tag{18}$$

where $\mathcal{A}_{\mathrm{wrong}} = \{\mathrm{zero}, \mathrm{shuffled}, \mathrm{reversed}\}$. This measures action dependence, not semantic correctness. MAE is computed over future RGB pixels in 0–255 pixel units.

Table 3. Design trajectory. Early methods explain why the final objective separates low-frequency control from high-frequency rendering.

Stage	Attempt	Outcome
Visual LoRA	Train only no-action visual continuation.	Strong visual baseline, but exactly action-blind: $S_{\mathrm{rgb}} = 0.000$.
Global/temporal action tokens	Encode actions as conditioning tokens exposed broadly to the transformer.	Could change outputs, but action gradients interfered with rendering and often produced blur or wobble.
AdaLN action modulation	Modulate hidden states with action-derived shifts/scales.	On a 512-window diagnostic subset, PSNR rose to 19.539 and SSIM to 0.795, but sharpness collapsed to 0.120 and motion to 0.511.
Teacher-preserving bottleneck	Freeze the visual model and train only action modules.	Reduced visual damage, but weak objectives could still be ignored or trade off motion and detail.
V4-LF	Full 112D actions, latent-time bottleneck, low-frequency target/delta supervision, high-frequency teacher preservation.	Full-val action sensitivity reached $S_{\mathrm{rgb}} = 2.262$ while PSNR/SSIM/FVD remained near the no-action baseline.

To test semantic alignment, we compare correct actions against wrong actions relative to the recorded future:

$$\begin{aligned} \Delta\mathrm{PSNR} = {} & \mathrm{PSNR}(G(a_c), x_{\mathrm{gt}}) \\ & - \frac{1}{3} \sum_{a' \in \mathcal{A}_{\mathrm{wrong}}} \mathrm{PSNR}(G(a'), x_{\mathrm{gt}}). \end{aligned} \tag{19}$$

Positive ΔPSNR or ΔSSIM means the correct action sequence is closer to the actual future than wrong actions.

6. Experiments

6.1. Full-validation comparison: preserving visual quality while adding action sensitivity

The final validation campaign evaluates the no-action baseline and the selected V4-LF rank-64 checkpoint on all 1,904 validation windows. For the action model, we generate four modes per window: correct, zero, shuffled, and reversed-future actions.

The modes are defined as follows. **No-action** is the visual LoRA baseline trained on the same future-generation task but given no ego-action input; its output is independent of the action sequence. **V4-LF correct** uses the recorded ego-action sequence aligned with the validation clip. **V4-LF zero** replaces the normalized action tensor with zeros, corresponding to the training-set mean action in normalized coordinates. **V4-LF shuffled** uses an action sequence from a different validation clip. **V4-LF reversed** keeps the context actions fixed but reverses the future action sequence in time. These counterfactual modes test whether the generated future depends on the correct action sequence rather than only on visual context.

The full-validation comparison demonstrates the intended tradeoff: V4-LF introduces counterfactual action

Table 4. Full 1,904-window validation. The action model adds controllability while remaining close to the no-action visual baseline. FVD is lower better.

Model/mode	$\mathrm{S_{rgb}}$	FVD	PSNR	SSIM
No-action	0.000	**11.702**	17.153	0.70789
V4-LF correct	**2.262**	11.747	17.180	**0.70816**
V4-LF zero	–	11.781	17.184	0.70803
V4-LF shuffled	–	11.761	**17.186**	0.70802
V4-LF reversed	–	11.761	17.182	0.70800

Figure 2. Full validation across no-action and V4 action modes. The selected action model is compared not only to no-action, but also to wrong-action generations. Correct actions are the best V4 mode by FVD-style distance while remaining visually close to no-action.

sensitivity without sacrificing the pretrained model's visual fidelity. The no-action baseline is action-blind by construction, with $S_{\mathrm{rgb}} = 0.000$, while the selected action model reaches $S_{\mathrm{rgb}} = 2.262$ across all 1,904 validation windows. At the same time, visual quality remains near baseline: PSNR changes from 17.153 to 17.180, SSIM from 0.70789 to 0.70816, and FVD-style distance from 11.702 to 11.747. This shows that the action pathway changes the generated future under counterfactual actions while preserving the visual behavior of the strong no-action predictor.

6.2. Correct actions versus wrong actions

Action sensitivity alone could mean the model changes arbitrarily when actions change. We therefore stratify validation windows by action magnitude. The high-action stratum is the union of braking, acceleration, and turning windows. Table 5 shows the strongest semantic result: on high-action clips, correct actions are slightly closer to the recorded future than wrong actions.

The effect is small, but the sign is important. It is strongest on acceleration and turning, where future ego actions should be most informative. Braking remains weak, and low-action clips are not the right controllability test because the future is already strongly determined by visual context.

Table 5. Correct-vs-wrong advantage by stratum. Positive ΔPSNR/ΔSSIM means correct actions are closer to ground truth than wrong actions.

Stratum	clips	$\mathrm{S_{rgb}}$	ΔPSNR	ΔSSIM
All	1904	2.262	-0.00419	+0.00015
High action	1055	2.303	+0.00192	+0.00027
Accelerate	408	2.484	+0.00493	+0.00069
Turning	509	**2.547**	+0.00419	+0.00054
Brake	402	2.041	-0.00064	-0.00003
Low action	571	2.335	-0.01786	-0.00001

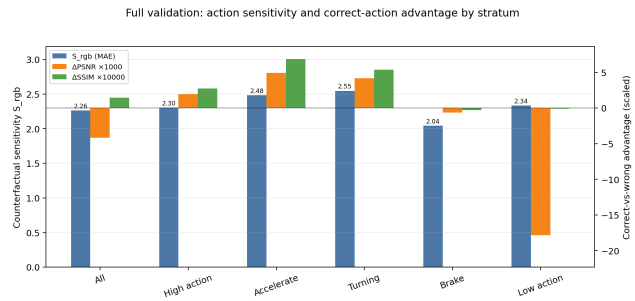

Figure 3. Action sensitivity and correct-action advantage by stratum. The model is most sensitive on turning and acceleration clips, and those same strata show positive correct-vs-wrong PSNR/SSIM advantage.

Table 6. 512-window diagnostic subset. AdaLN improves PSNR/SSIM but collapses sharpness and motion.

Model	PSNR	SSIM	Sharp.	Motion
No-action	16.893	0.695	0.250	**1.262**
Frame AdaLN	**19.539**	**0.795**	0.120	0.511
HF-teacher v2	17.970	0.731	0.140	0.927
V4 action-strong	17.803	0.724	0.164	0.995
V4 main text	16.844	0.691	0.245	1.216

6.3. Why naive action conditioning was insufficient

Table 6 and Fig. 4 summarize the core diagnostic that changed the project direction. AdaLN-style action conditioning achieved the best PSNR and SSIM on the 512-window diagnostic subset, but it did so by collapsing sharpness and motion. This is exactly the failure mode we wanted to avoid.

6.4. Capacity and checkpoint behavior

We also tested whether dense 112D action conditioning was limited by adaptation capacity. Rank 32 briefly reached higher sensitivity, but was unstable. Rank 64 gave the selected tradeoff. Rank 128 was not a monotonic improvement.

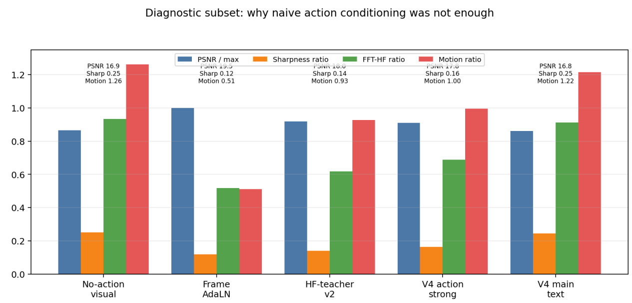

Figure 4. Diagnostic subset comparison. The high-PSNR AdaLN model loses high-frequency detail and motion. The final V4 direction preserves these quantities closer to the no-action visual model.

7. Discussion

What improved? The cleanest improvement is controllability. The no-action model has $S_{\mathrm{rgb}} = 0.000$ by construction. The final model reaches $S_{\mathrm{rgb}} = 2.262$ on full validation and $S_{\mathrm{rgb}} = 2.303$ on high-action clips. It does this while staying visually close to no-action: FVD changes from 11.702 to 11.747, PSNR from 17.153 to 17.180, and SSIM from 0.70789 to 0.70816.

Why is the result not just noise? The final validation includes 1,904 windows and compares correct actions against three wrong-action modes. Correct actions are not globally better on every window, but they become positive on the high-action subset and especially on acceleration and turning. This is the expected direction if actions matter primarily when the future is not determined by context alone.

Interpretation and remaining limitations. The correct-vs-wrong advantage is still modest, but its distribution is meaningful. The strongest positive alignment appears in acceleration and turning clips, where future ego-motion visibly changes the scene trajectory and should be informative beyond visual context alone. Braking is a harder case in this dataset. In Waymo clips, braking is often smooth, controlled, and visually subtle rather than abrupt, unlike many human-driving examples. Visual inspection shows that deceleration frequently changes the future through small shifts in forward progress rather than obvious scene-level events, making it harder for pixel and structural metrics to separate correct braking actions from nearby counterfactuals. Thus, the result should be interpreted as a transition from action-blind prediction to measurable action sensitivity and early semantic alignment, with braking alignment left as a more difficult future direction.

Why the finding matters. The main contribution is not that V4-LF beats the no-action model on every visual metric; that is not the goal. The goal is to add action dependence without destroying the pretrained video prior. Across the design trajectory, we found a consistent pattern: broad or unconstrained action injection can make the model react to actions, but often by damaging rendering quality, smoothing motion, or collapsing high-frequency detail. The final V4-LF design gives a cleaner tradeoff: low-frequency temporal conditioning controls coarse future evolution, while high-frequency teacher preservation protects appearance, texture, and local visual structure. This supports the central claim that action control and visual fidelity should be treated as separate objectives in pretrained video diffusion transformers, and it is consistent with recent work emphasizing temporal consistency, feature alignment, and structured multi-condition control in video diffusion [10, 9, 20].

8. Conclusion

We studied action conditioning for pretrained diffusion-transformer driving video generation. The main finding is that action control and visual fidelity are separate axes: a no-action model can be a strong visual predictor while remaining action-blind by construction. Naive action conditioning exposes this tension, since some mechanisms improve pixel metrics by smoothing motion or suppressing high-frequency detail rather than learning useful control. Our final low-frequency action-conditioned model addresses this by treating ego-actions as coarse temporal control signals while preserving the pretrained visual prior. On full validation, it increases counterfactual sensitivity from $S_{\mathrm{rgb}} = 0.000$ to $S_{\mathrm{rgb}} = 2.262$ while keeping PSNR, SSIM, and FVD-style distance near the no-action baseline. On high-action clips, especially acceleration and turning, correct actions are closer to the recorded future than wrong action sequences. These results show that low-frequency action conditioning can convert an action-blind driving-video predictor into an action-sensitive generator without visual collapse, establishing a practical direction for controllable diffusion-transformer world models.

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